
Deep Generative Models

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Deep generative models [1] have advanced data generation and modification by learning underlying data distributions and producing new unseen samples that resemble the original dataset. This capability has significant implications across various fields, including art, medical image synthesis, and scientific simulations. The importance of deep generative models lies in their ability to: (1) Synthesize data when real-world data is scarce, useful for training machine learning models; (2) Enhance quality and creativity, enabling machines to produce art and designs autonomously; (3) Improve data augmentation, crucial for training robust models without overfitting; (4) Bridge modalities, converting between images and other types of data, such as text-to-image models. In recent years, three main types of generative models have gained prominence, including Generative Adversarial Networks (GANs) [2], Variational Autoencoders (VAEs) [3], and Denoising Diffusion Probabilistic Models (DDPMs) [4]. These references can be used as the main reading materials. The following detailed references are recommended for further advancements in each category.

1 From GANs to DDPMs

GANs introduced adversarial learning, where a generator learns to produce realistic samples by competing with a discriminator that distinguishes generated from real data. GANs can produce sharp, high-quality images, but their adversarial optimization is difficult to stabilize and may lead to mode collapse, where only part of the data distribution is represented. VAEs instead formulate generation through probabilistic latent-variable modeling: an encoder maps data to a latent distribution and a decoder reconstructs or generates samples from this space. VAEs provide a structured and interpretable latent representation, although conventional likelihood objectives can result in smoother outputs. DDPMs take a different approach by gradually corrupting data with Gaussian noise and learning the reverse denoising process that transforms noise back into data. Their simple denoising objective generally provides more stable training while achieving high sample quality and diversity, at the cost of computationally expensive iterative sampling. The progression from GANs and VAEs to DDPMs therefore illustrates a central trade-off in generative modeling between sample quality, diversity, optimization stability, and generation speed.

2 From DDPMs to Flow Matching

Research following DDPMs has increasingly focused on understanding and accelerating the trajectory that connects noise and data. Improved DDPMs [5] refine noise schedules and reverse-process variance to reduce sampling cost, while Denoising Diffusion Implicit Models (DDIMs) [6] show that generation can follow deterministic, non-Markovian trajectories and require substantially fewer sampling steps. Score-based generative models [7] generalize diffusion to continuous time using stochastic differential equations (SDEs), where a neural network estimates the score of noisy data distributions, and generation can be performed either through a reverse-time SDE or an equivalent deterministic probability-flow ordinary differential equation (ODE). Flow Matching [8] simplifies this perspective further by directly learning the velocity field that transports samples from a simple noise distribution toward the data distribution, rather than first estimating a score and deriving the corresponding sampler. Related approaches such as rectified flows [9] encourage straighter trans-

port trajectories, allowing accurate generation with fewer numerical integration steps. Thus, the development from DDPMs to score-based models and Flow Matching can be viewed as a shift from learning discrete denoising transitions toward directly learning continuous and increasingly efficient transport between noise and data.

3 From Generation to Editing and Segmentation

Modern generative models have evolved from unconditional sample generation into flexible conditional models that can manipulate, interpret, and analyze existing data. Guided diffusion [10] controls the reverse process using additional information, with classifier guidance or classifier-free guidance [11] allowing attributes, classes, or text prompts to influence the generated content. Latent diffusion models [12], including Stable Diffusion, perform this process in a compressed latent space rather than directly in pixel space, substantially reducing computational cost while enabling high-resolution generation. Conditioning mechanisms such as cross-attention connect textual or other contextual information to spatial image representations and enable applications such as prompt-based editing [13], where selected concepts can be modified while much of the original image remains unchanged. The same generative principles can also be adapted to discriminative tasks: diffusion and latent-diffusion models have been used to generate segmentation masks [14] or provide counterfactual explanations [15]. More recent foundation models further unify generation and editing within a single flow-based or diffusion-based framework, illustrating a broader transition from generative models as image synthesizers toward general-purpose models for conditional generation, manipulation, interpretation, and structured prediction.

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